

QUESTION 1: TRUE/FALSE QUESTIONS (TOTAL, 10 MARKS)

- 1.(1 point) The probability of measuring the quantum state $|+\rangle$ with the projector $|0\rangle\langle 0|$, is $1/2$
- 2.(1 point) For classical-quantum state ρ_{AB} , we always have $P_{\text{guess}}(X|B) \geq P_{\text{guess}}(X)$
- 3.(1 point) The Pauli-Y matrix is both Hermitian and unitary.
4. (1 point) Just because a quantum teleportation scheme works for any qubit state $|\varphi\rangle$, this does not mean that it can also teleport a state that is entangled with a third party.
5. (1 point) For pure state with density matrix p , we have $\text{Tr}(p^2) \leq 1$
- 6.(1 point) In BB84 QKD protocol, Bob will immediately see that an eavesdropper was interfering with the communication. He does not need to communicate with Alice to realise this.
- 7.(1 point) $p = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is a valid density matrix.
- 8.(1 point) Trace distance of two states, p and a , is a measure related to the maximum probability of successfully distinguishing between these two states.
- 9.(1 point) Quantum entanglement can be used for superluminal communication.
- 10.(1 point) If system A is maximally entangled with system B, it can be simultaneously entangled with system C.

QUESTION 2(TOTAL 10 POINTS)

Alice and Bob share the state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

1. (2 points) Express this state in terms of the eigenstates of the Pauli X operator, i.e., write it in terms of the basis states $|+\rangle$ and $|-\rangle$
2. (3 points) Give the probabilities of all possible outcome pairs in case Alice and Bob either both measure in the computational or both in the Hadamard basis.

Assume now instead that Alice and Bob share the state $|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

3. (2 points) Also express this state in terms of the eigenstates of the Pauli X operator.
4. (3 points) Give the probabilities of all possible outcome pairs in case Alice and Bob either both measure in the computational or both measure in the Hadamard basis.

QUESTION 3(TOTAL 10 POINTS)

Given matrix $p = \frac{1}{3}|0\rangle\langle 0| + \frac{2}{3}|+\rangle\langle +|$. Give its matrix form in the standard basis, $\text{Tr}(p)$ and $\text{Tr}(p^2)$

QUESTION 4(TOTAL 10 POINTS)

Suppose we have a qubit in the state $|0\rangle$ and we measure the observable X (Pauli matrix). What is the average value of X? What is the standard deviation of X?

QUESTION 5(TOTAL 10 POINTS)

Consider quantum states: $|\Phi 1\rangle = |0\rangle$, $|\Phi 2\rangle = |1\rangle$, $|\Phi 3\rangle = |+\rangle$, $|\Phi 4\rangle = |-\rangle$,

$|\Phi 5\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|\Phi 6\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Consider now the following game: First, Alice throw a fair die and notes down the outcome as $x \in \{1, 2, 3, 4, 5, 6\}$. Then, Alice prepares the state $|\Phi x\rangle$ depending on the outcome x and sends it to

Bob. After this, Alice will keep X and Bob will hold a system B with the state Alice prepared.

1. (3 points) Write down the classical-quantum state P_{XB} shared between Alice and Bob that describes the above situation.
2. (3 points) Is P_{XB} entangled?
3. (4 points) What is $P_{\text{guess}}(X)$?

QUESTION 6 (TOTAL 10 POINTS)

- 1.(5 points) What is the trace distance $D(p_0, p_1)$ between $p_0 = \begin{pmatrix} \frac{1}{3} & \frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} & \frac{2}{3} \end{pmatrix}$ and $p_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
- 2.(5 points) What is the optimal measurement operator that distinguishes p_0 and p_1 ?